CSE 310: Computer Networks
Homework #1
Stony Brook University

Total: 36 pts
Rubrics: as shown below

1- (10 pts) A **1200 Mbit file** is segmented into **800 packets**. The path from source to destination has **4 hops** (links). Link bandwidth is **100 Mbps** for the first 3 links, but the last link is only **50 Mbps**. What is the total transmission time?

2- (8 pts, 4 each) Consider a network with 3 nodes (A, B, C) connected by two routers in a chain: A ↔ Router1 ↔ Router2 ↔ B. Node C is also connected to Router2.

All links have:

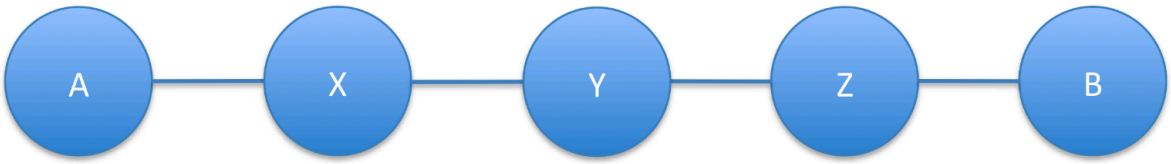
- Transmission rate: **100 Mbps**
- Propagation delay: **5 ms per link**
- Can handle **15 simultaneous connections** in circuit switching mode

Scenario 1:
If A and B establish a circuit-switched connection for a 4K video stream that lasts 2 hours, what is their **maximum sustainable throughput**?

Scenario 2:
In packet-switched mode, if A→B and C→B both start large file transfers simultaneously when the network is otherwise idle:

- What is the **maximum rate** for A→B?
- What is the **maximum rate** for C→B?

3- (8 pts) Assume we have the following network where each link is 100 m long and has a transmission rate of 1Gbps. Assume that the propagation speed is 2 * 10^8 m/s and all packets have the same size of 1000 B (bytes).



a. How long will a packet sent from A to get to B take?
b. Assume host A generates 50 packets instantly. How long will the first packet wait in A's queue before being transmitted? How long will the 50th packet wait in A's queue before being transmitted? You can assume that no other packets are sent.
c. Suppose now that host A closed the previous connection and started sending 300 packets every 0.1s. What is the throughput it achieves, assuming no packet losses?
d. If the transmission rate of the link between Y and Z changed to 10 Mbps. What is the maximum throughput that host A can achieve? If A tries to transmit packets at a higher rate, what will happen? Explain.

4- **(10 pts) Wireshark Exercise**:

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures ("sniffs") messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure 1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally runs on your computer. The packet sniffer, shown within the dashed rectangle in Figure 1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. Recall from the discussion from section 1.5 in the text that messages exchanged by higher layer protocols such as HTTP, FTP, TCP, UDP, DNS, or IP all are eventually encapsulated in link-layer frames that are transmitted over physical media such as an Ethernet cable. In Figure 1, the assumed physical media is an Ethernet, and so all upper layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must "understand" the structure of all messages exchanged by protocols. For example, suppose we are interested in displaying the various fields in messages exchanged by the

HTTP protocol in Figure 1. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string "GET," "POST," or "HEAD".

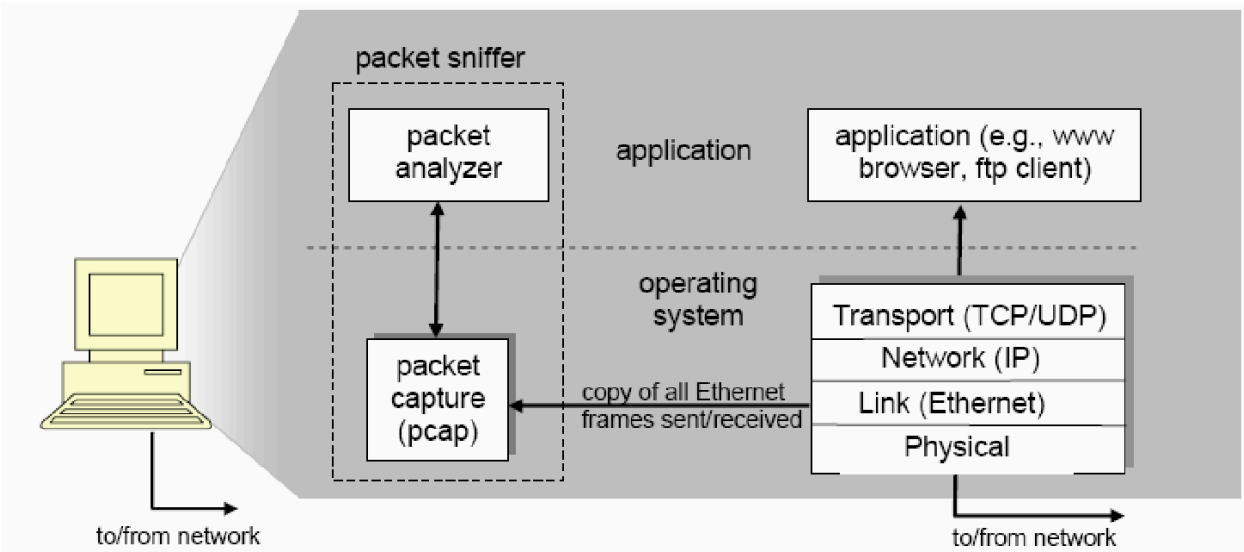


Figure 1: Packet sniffer structure

We will be using the Wireshark packet sniffer [http://www.wireshark.org/] for our homework, allowing us to display the contents of messages being sent/received from/by protocols at different levels of the protocol stack.

Running Wireshark

When you run the Wireshark program, the Wireshark graphical user interface shown in Figure 2 will be displayed. Initially, no data will be displayed in the various windows.

The Wireshark interface has five major components:

- The **command menus** are standard pulldown menus located at the top of the window. Of interest to us now are the File and Capture menus. The File menu allows you to save captured packet data or open a file containing previously captured packet data and exit the Wireshark application. The Capture menu allows you to begin packet capture.

- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number (assigned by Wireshark; this is *not* a packet number contained in any protocol's header), the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific

information contained in the packet. The packet listing can be sorted according to any of these categories by clicking on a column name. The protocol type field lists the highest-level protocol that sent or received this packet, i.e., the protocol that is the source or ultimate sink for this packet.

- The **packet-header details window** provides details about the packet selected (highlighted) in the packet listing window. (To select a packet in the packet listing window, place the cursor over the packet's one-line summary in the packet listing window and click with the left mouse button.). These details include information about the Ethernet frame (assuming the packet was sent/received over an Ethernet interface) and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus-or-minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.

- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

- Towards the top of the Wireshark graphical user interface, is the **packet display filter field,** into which a protocol name or other information can be entered in order to *filter* the information displayed in the packet-listing window (and hence the packet-header and packet-contents windows). In the example below, we'll use the packet-display filter field to have Wireshark hide (not display) packets except those that correspond to HTTP messages.

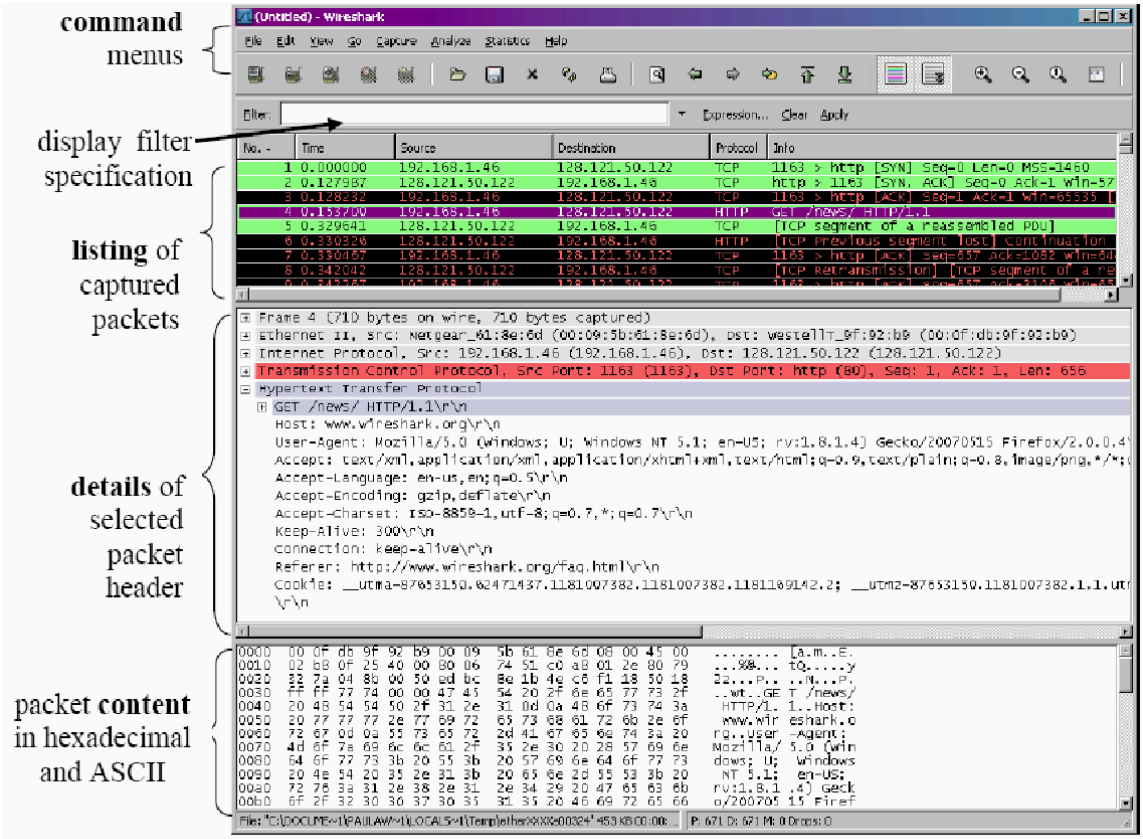


Figure 2: Wireshark Graphical User Interface

Answer the following questions:

Open the provided simulated trace file (introduction.pcap under Homework #1 on Brightspace) with Wireshark, and answer the following questions based on your Wireshark experimentation:

a. List 10 different protocols that appear in the protocol column in the unfiltered packet-listing window.

b. Try the filter function: type "http", then press the "Apply" button. How long did it take from when the first HTTP GET message was sent until the first HTTP OK reply was received? (By default, the value of the Time column in the packet listing window is the amount of time in seconds, since Wireshark tracing began. To display the Time field in time-of-day format, select the Wireshark *View* pull down menu, then select Time *Display Format*, then select *Time-of-day*.)

c. The HTTP server being used is named gaia.cs.umass.edu. What was the Internet address (IP) of the HTTP server? What was the port number of the server? What was the IP address of the client? What was the port number for the client? You may need to click the + sign to expand the information in the detailed packet content window.

d. There are two HTTP GET request messages. The first one was successful in getting some text data back from the server. What was the line of text data that was received?

e. Refer to the five layers of the Internet protocol stack: application, transport, network, link, and physical. Write down which protocol was used, respectively, in the first four layers (ignoring the physical layer) with the HTTP server?